

Second Steven

Everything you've told me, organised so you never have to say it twice.

Compiled	1 September 2026
Source	Every message in the build conversation, verbatim
Repository	github.com/Stevenfrancis2/hey
Branch	<code>claude/telegram-personal-assistant-vodi6o</code>
Also lives at	<code>docs/</code> in the repo, and the nightly Telegram archive

How to use this

If this project ever has to restart — a new session, a new machine, a different assistant, or six months from now when the detail has gone — hand over this file. It carries the whole context: who you are, what you're running, every decision made and why, what's built, and your own words verbatim at the back.

Nothing in here needs to be brainstormed again.

1 · Who Steven is

Based in Lebanon. Runs several things at once and thinks in fragments — ideas arrive mid-task, hands full, and are worth about ninety seconds before they're gone.

Lebanon shapes almost every practical answer: parts and machines often aren't available locally, importing is the normal path, and shipping cost and lead time matter as much as price. Never assume a local supplier exists.

Background

Engineering — public repositories include an autonomous car, a blind stick, Arduino work, an accent detector, and (December 2025) **ragagent** and **ragagent1**: retrieval-augmented agent work. That last one matters — it's the same shape as the bank project and as this system's memory.

2 · What he's running

Room	What it is	Weight
Cligli	3D printing and assembly business. Products, orders, suppliers, print farm, filament. Wants more sales and better suppliers.	Heavy
Bank AI	Side project with a friend building AI agents for banks. Friend owns the platform; Steven deploys the local LLM.	Deadline
Drones / FPV	Building, repairing, training, DCL racing, content.	Heavy
Remote work	The day job — automation. Client projects with deadlines and their own material and data.	Medium
Royal Pizza	His father's business. Helps with the dough.	Keep light — his instruction
Land	600 sqm plot. Guesthouse, own house, or hold.	Open question
Finance	The books across every business. Starting to invest.	High value
Body	Gym, training, food, calories.	Small

3 · The bank AI project

The only thing here with a date on it, and the only one where someone else is depending on him. It outranks everything.

What it is

- Building AI agents for banks, with a friend.
- The friend owns the platform. Steven's part: deploying a **local LLM**, **Docker**, and **L1 / L2 agent tiers**.
- Confirmed: **NVIDIA NeMo** (the framework, not Mistral's model).
- Roughly one month of study needed, from 1 September 2026.
- Wants a shared space — Google Drive or similar — where either of them can drop anything.

What that means for the study plan

NVIDIA NeMo makes this infrastructure-led rather than model science: containerised GPU inference, NIM microservices, retrieval, and — because it's a bank — **guardrails and evaluation**. That last part is what most people skip and exactly what a bank will audit.

Start from ragagent, not from zero.

The December 2025 retrieval-agent work is the same shape. Reading it first is worth days.

4 · The fleet

Quads

Aircraft	Class	Wind limit	Gust limit
iFlight Nazgûl F5 V3	5-inch freestyle	32 km/h	38 km/h
Flywoo DarkStar 22	2.2" micro long range	16 km/h	22 km/h
BetaFPV Meteor 75	75 mm whoop	8 km/h	12 km/h

These limits drive the flyability answers and are editable. They matter because the three aircraft are not comparable: on a real forecast day the 5-inch got 19 flyable hours, the micro 11, and the whoop 2. One weather score for all three would be useless.

Radios

- RadioMaster TX16S MAX
- RadioMaster Boxer

Printers

- Bambu Lab A1 × 7 — the farm
- Bambu Lab H2C × 1

Wants to be kept updated on firmware releases and known glitches for these.

Assumptions to confirm

- "tx15 max" read as **TX16S MAX** — there is no TX15.
- DarkStar 22 brand assumed **Flywoo**.
- "h2c" read as Bambu Lab **H2C** — confirm the exact model.
- Batteries not yet given — he said he'd supply them.

5 · Decisions made, and why

Locked

Decision	Choice	Reasoning
Language	TypeScript, Node 22	grammY is the best Telegram library; one language across bot, jobs, API and console
Hosting	Fly.io + Neon Postgres	~\$5–30/mo, one command to deploy; migratable to own hardware later
Voice notes	Yes, day one	Hands are usually full. Groq Whisper, ~\$2/mo
Calendar	Google Calendar	Read and write; lands in Phase 3
Embeddings	Voyage voyage-3.5-lite	Anthropic has no embeddings endpoint; Voyage is who they recommend
Database	Postgres + pgvector	Relational, vectors, graph edges and the job queue in one store
Graph	edges table, recursive CTE	No Neo4j — real cost, no benefit at this size
Jobs	pg-boss	No Redis. Cron + delayed jobs with ACID
Weather	Open-Meteo	Free, no key, hourly gusts
Timezone	Asia/Beirut	Lebanon

Telegram or a UI? — both, with a split

His question: *the farm manager is a UI full of things and this has twenty times more — so is Telegram right at all?*

The answer isn't chat versus UI. It's which direction the information is moving. Chat wins for unstructured input while moving, and for one fact coming back. A UI wins for anything dense or comparative — thirty print jobs, a cash position, a calendar. A print queue is a table: you scan it and spot the stalled row. Nobody can scan a table read out to them.

So more features don't mean more chat. The chat box stays the same size no matter how many rooms exist. The UI is where the growth goes.

And the failure mode he was sensing: a chat-only system gets *worse* as it grows, because discovery collapses. With three features you remember the commands; with sixty you use the same five and the rest may as well not exist. A UI is self-documenting.

Telegram is the microphone. The console is the cockpit.

The Raspberry Pi farm manager stays on the Pi

He built a filament and print-farm manager earlier with Claude Code; it lives on a Raspberry Pi and is not on GitHub (checked — no repository in his account by that name). It should not be ported: **the printers are on his local network and a cloud container cannot reach them.** Instead the Pi keeps local control and gets a small outbound sync pushing job state and filament levels up. Its screens get embedded as the Cligli room.

That federation is what produces sentences like *"that job was underpriced"* — which the farm manager alone can never say, because it doesn't know what he charged.

6 · Everything asked for, by topic

Capture and memory

- One inbox for everything — ideas, tasks, links, photos, voice.
- Voice and chat both. Zero friction, no categorising.
- Graph memory — knows how everything connects.
- Should know everything he knows: drones, 3D printing, marketing, social media, restaurants, stocks, crypto.
- Attach other people's messages and voice notes instead of re-explaining.

Organisation — his stated top priority

- "Everything organised where it should be."
- Compartments that stand alone, plus one shared place to throw things at.
- Projects with deadlines, and each project's material and data attached.

Durability — insisted on twice

- Store all the chat, structured, in a file.
- "If something bad happens I don't need to brainstorm again."
- A PDF with everything, organised.

Finance

- Manage savings, billing, what he can and cannot buy.
- Report in plain language: "we got this, we sold this, we bought this" — for all businesses.
- ~4k from one retail store — confirmed already paid, so it books as revenue.

Investing

- Starting to invest from salary.
- Watch drone company stocks, nuclear, water, AI.
- Investment opportunities in the news.
- Crypto — good time to buy, which coins, which projects.

Built to research and track his thesis, not to recommend trades — it can't see his full position, and a confident call from a system he trusts would be worse than useless.

Sourcing and suppliers

- Knows the brand and part; not available in Lebanon.
- Find it — Amazon or elsewhere — with full details and images.
- Example given: an oven part for Royal Pizza.
- More suppliers, and more sales online.
- May give access to Instagram Ads and the Cligli catalogue.

Meta's Ads API takes weeks to approve — request access early, build it late.

Flying

- "The only thing I need is if the weather is good to fly my drones."
- Per-aircraft, because the three quads are not comparable.
- Quick tips and technique from the bot.

3D printing

- Print farm and filament inventory feeding job costing.
- Bambu Lab firmware updates and known glitches.
- New machines worth buying.
- Trending 3D-printed products and the wider toy market.

Study

- He supplies the material; it works out how to learn it as fast as possible.
- Primarily for the bank project, but general.

Daily research

- AI news, stocks, crypto.
- Trending 3D-printed toys and the major toy market.
- How to improve Cligli, new machines, "new doors".

"New doors" assumed to mean new sales channels and openings — still unconfirmed.

Body

- Gym reminders, gym tips, calorie tracking.

Land

- 600 m² plot, recently acquired.
- Guesthouse or build his own house.
- Wants feedback, research and a market study.

Automation

- Find things it can automate on his behalf, or produce a plan to automate them.

Access

- Multi-device: laptop, PC, iPad, phone.
- A shared space with the bank-project friend.

Sharing is per-room and default-deny — the friend gets the bank room only, never finances or health.

7 · What's built and working

Phase	What	State
0	Capture loop — text, voice, photo, document, forwards. Instant reaction ack, Whisper transcription, hybrid search.	Built
1	The brain — nine rooms, Opus 5 with adaptive thinking, tasks, reminders, hosted web search, daily + weekly briefs.	Built
1b	Archive — nightly Markdown + JSON over Telegram, /export.	Built
1c	Projects with deadlines. Watchlist with thesis.	Built
1d	The console — every device, PWA, magic-link login.	Built
1e	Fleet registry and per-aircraft flyability.	Built
2	Study planner	Next
3	Finance ledger	After
4	Pi farm-manager sync	After

The design rule everything follows

The write path is dumb and instant. The read path is smart and slow. A message is one database insert plus a queued job, then it acknowledges. Transcription, classification and embedding happen seconds later in the background. He never waits on a model — which is the only reason this survives being used one-handed.

And: he never chooses between noting something and asking something. Everything is stored; it replies only when he actually asked. There is no mode to switch.

Commands

Command	Does
/fly	Can I fly — per quad, with windows
/gear	The fleet
/recall <query>	Search everything ever said
/tasks /projects	What's open; deadlines
/brief	Today's brief now
/export	Everything as files, right now
/login	Open the console on this device
/costs	What it's spending

8 · Still open

- "New doors" — sales channels, or doors as a product?
- Does the friend need write access to the shared room, or read only?
- Exact flying location — currently defaulting to Beirut coordinates.

- What Cligli is exactly — assumed printing and assembly of a physical product.
- Battery inventory — he said he'd supply it.
- Confirm TX16S MAX, DarkStar 22 brand, and the Bambu H2C model.
- The Pi farm-manager code, when he's home.

9 · His own words, verbatim

Kept exactly as sent, in order, typos and all. This is the primary source — everything above is derived from it. Where the two ever disagree, this wins.

Message 1 — the original brief

"Build a personal assistant linked to telegram where I send everything in my mind like what if an idea crossed my mind about one my businesses or. To do thing or everything like I want a 2nd brain with me anytime reminds me of things solves things for me web research stocks reminders calendar meeting like literally. Second brain maybe link it to. Graphical memory low like think bout what we can build I m running cligli t the same time mingling atefpv content ne drones nd building repairing trining for Dcl helping father in making the dough for royal pizza mingling cligli printing assembling working my remote I utomtion I want news about ai war wether id I ant to fly like literally. Second Steven smarter way more cpabilities md moyen duplicate ,y time nd Sven lot of my time why do your ink put. Plan first what frameworks what everything"

Message 2 — the expansion

"In addition I remembered in a month I have to learn a lot nd lots of stuff because I took a side project with my friend buildignngents for banks he has the platform I will be deploying it local llm Nemo docker l1 l2 agent ,aube have. Google Drive or so,e sharpening where I can drop info bout lieerqly everything and he has ccess to those everything's you know? ... I want everything one place of we can make it a web app for phone or maybe an app like idk you think bout it I want easibility structure like each compartment alone maybe nd one share dplace we I throw things at the agent ... I want to get more sales whether online or get to know more supplier like I have around 4k from one retail store I sold stuff to like it maybe help me get finance right ... Study organized or planner I give the material he knows how I can earn them s fast s ossible ... this is why iam brainstorming with you for the bot"

Message 3 — on the interface (answering the four questions)

"I built it with Claude code earlier it is on my raspberry pi when I am back home I can share the code with you so you add it but u think the best way is telegram or a ui with many things like the fame manager is. I with lots of stuff this agent has 20x more things so think about it"

Message 4 — remote work, durability, investing

"Maybe we addd for remote work I maybe talk about projects and deadlines I also share each projects material and data maybe I need something in so the assistant how to fix it how to build it but organization is my top thing I want everything organized where it should be and as mentioned I want you to store allllt he chat I send like each e,saga sent structure those and put them in a file if something bad happen I don't need to brainstorm gain. Will get into stocks soon from my salary and stuff keep eye on drone company stocks nuclear water ai investment opportunities in the news thing or crypto good time to buy maybe what coins what projects"

Message 5 — Lebanon, sourcing, multi-device

"I am reading what astounding r talking I agree there is plenty of things bout cligli the restaurant don't stress it a lot ok for example to have an oven I nknow th ebbrand the parts ar Neto available in Lebanon maybe the bro can find them on Amazon with all details and images tâches you know? I can attach other people. Lice's instead of explaining this is still rbaistorming dn adding to the bot I was thinking if I am siting on th e desktop how can I check the up? Work on that on laptop and pc and iPad and phone compatibility and access"

Message 6 — the fleet

"U also mentioned something in cost about fly .io only thing I need is if the weather is good to fly my drones I have 5 inch Nazgûl f5 v3 iflifght I have meteor 75 betafpv and dark star 22 will also tell batteries and maybe I need quick tips or stuff from the bot i have 2 controllers radio master tx15 max and radio master boxer I have 7 Bambi lab a1 keep upadtaed on updates and glitches and h2c Bambi lab 1"

Message 7 — where to work

"wasnt me to poen it in vscode and claude code ther and continue or hereis good? if yes guide me what do you weqant m,e to do and i insinst on the pdf having all the chat io sent i want them organized so wehenevr i ned to build or somehtign happen no need to repeat my self"

End of dossier. Regenerate any time — the live version is the nightly archive delivered over Telegram, and docs/ in the repository.